

# DOCTORAGENT-RL: A MULTI-AGENT COLLABORATIVE REINFORCEMENT LEARNING SYSTEM FOR MULTI-TURN CLINICAL DIALOGUE

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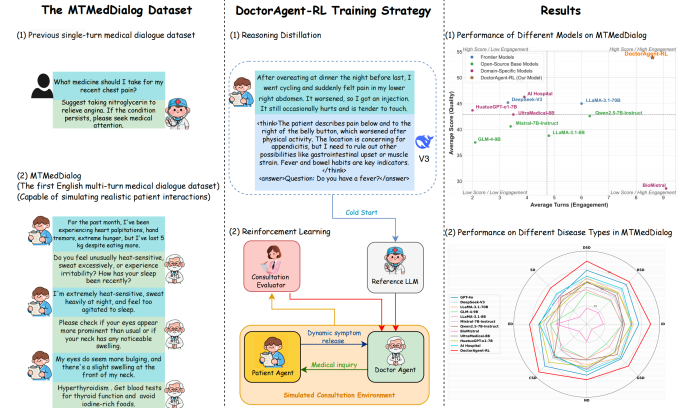
## ABSTRACT

Large language models (LLMs) excel at biomedical question answering but struggle in real clinical consultations. Single-round systems require patients to list all symptoms initially, often causing vague diagnoses. Traditional multi-turn models, limited by static supervised learning, lack flexibility and cannot intelligently gather key clinical data. To overcome this, we propose DoctorAgent-RL, a multi-agent reinforcement learning (RL) framework that treats medical consultations as dynamic decision-making under uncertainty. The doctor agent optimizes its questioning strategy via multi-turn interactions with a patient agent, dynamically adjusting its information collection based on rewards from a Consultation Evaluator. This RL fine-tuning allows LLMs to develop clinical reasoning strategies, not just mimic existing dialogues. We also built MTMedDialog, a new English multi-turn medical dataset designed for interactive simulation. It contains detailed case profiles that allow a patient agent to progressively reveal symptoms in response to the doctor’s questions, offering a more realistic training and testing setting than static datasets. Experiments show DoctorAgent-RL outperforms existing models in diagnostic accuracy. This approach reduces the risk of misdiagnosis in time-sensitive situations, frees clinicians to focus on complex cases, and helps optimize the use of medical resources.

**Index Terms**— Medical Dialogue, Large Language Model, Reinforcement Learning, Multi-Agent System

## 1. INTRODUCTION

Large language models (LLMs) like ChatGPT [1] and LLaMA [2] show significant potential for assisting in clinical decision-making [3]. This capability is urgently needed in the context of a global doctor shortage and high-pressure healthcare



**Fig. 1.** Overview of this work. DoctorAgent-RL achieves state-of-the-art on the MTMedDialog task.

environments [4]. In these settings, time pressure on clinicians often leads to rapid diagnoses based on incomplete information, increasing misdiagnosis risks [5], while routine consultations consume resources that could be allocated to more critical cases [6]. Therefore, developing an intelligent system to improve primary care efficiency holds significant practical importance for optimizing resource allocation and alleviating healthcare strain [7].

Existing systems suffer from two fundamental limitations. First, the “answer-only” characteristic of single-turn systems [8] requires patients to describe their condition comprehensively at once, which conflicts with the clinical reality that complex conditions often necessitate multi-turn inquiries for clarification [9]. Second, current multi-turn dialogue methods that rely on fine-tuning are constrained by imitating static data patterns, and thus lack the dynamic decision-making mechanism required to reason like a clinician [10].

To address these challenges, we propose DoctorAgent-RL—a multi-agent reinforcement learning framework that formulates clinical reasoning as a Markov Decision Process (MDP). As illustrated in Figure 1, the framework consists of: (1) an LLM-based patient agent that generates pathologically

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<https://github.com/JarvisUSTC/DoctorAgent-RL>

consistent responses while mimicking real-world communication diversity; (2) a doctor agent initialized by cloning medical behaviors from real consultations and refined via reinforcement learning to develop effective questioning strategies; and (3) a consultation evaluator that provides multi-dimensional rewards based on diagnostic accuracy, information responsiveness, and question standardization. While existing datasets offer valuable multi-turn medical dialogues, they remain static and limit interactive training. To overcome this, we construct MTMedDialog—the first English dataset designed for dynamic, state-grounded patient simulation. Each dialogue is linked to a comprehensive patient profile, enabling realistic, incremental symptom disclosure aligned with the diagnostic process. Through reward-driven policy learning, DoctorAgent-RL enables the doctor agent to optimize diagnostic reasoning by dynamically adapting its questioning strategy during multi-turn interactions with a patient agent, allowing LLMs to develop clinically logical strategies beyond superficial imitation. Experiments show that DoctorAgent-RL outperforms existing models in diagnostic accuracy, demonstrating practical potential for assisting clinical consultations. This work offers a viable pathway toward intelligent diagnostic systems that can reduce misdiagnosis and improve healthcare resource allocation.

Our contributions are three-fold:

- We propose DoctorAgent-RL, a multi-agent RL framework where a doctor agent learns questioning strategies via reward-driven interactions with a patient agent, guided by a consultation evaluator.
- We introduce MTMedDialog, a novel multi-turn English medical dialogue dataset. It features hidden patient profiles that support dynamic symptom revelation across dialogue turns, enabling realistic simulation and evaluation of interactive diagnostic agents.
- DoctorAgent-RL achieves state-of-the-art performance in multi-turn medical consultations.

## 2. METHODOLOGY

### 2.1. Dataset creation process

We introduce MTMedDialog, a new benchmark designed for training and evaluating LLM through multi-turn medical consultations. Unlike previous datasets, each dialogue in MTMedDialog is grounded in a hidden patient profile, enabling a patient simulator to dynamically disclose symptoms and medical history in response to a doctor agent’s inquiries—rather than following a fixed script. This design facilitates the development and evaluation of agents with advanced, adaptive inquiry strategies. The dataset consists of 8,086 training and 2,082 test samples, curated from three Chinese medical dialogue benchmarks: IMCS21 [11], CHIP-MDCFNPC [12], and MedDG [13]. The test set spans 8 major disease categories, such as respiratory, infectious, and digestive system diseases.

As the source data originates from real doctor-patient conversations, we applied a two-stage denoising process: (1) filtering out dialogues with fewer than three turns via exact turn-count matching, and (2) removing noisy segments with consecutive meaningless responses using DeepSeek-V3 [14]. Throughout cleaning, we adhered to the original data splits to maintain evaluation reliability. Retained dialogues were translated into English using DeepSeek-V3, and the test set underwent manual review and refinement. We will release both Chinese and English versions of the test set.

### 2.2. Task Formulation

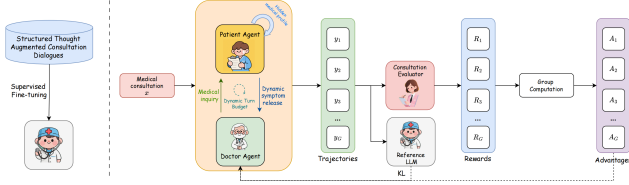
We model the medical consultation process as a collaborative, multi-agent RL system. A doctor agent optimizes its strategy through interaction with a patient agent, while a consultation evaluator acts as an arbiter, providing rewards to guide the doctor agent’s policy optimization.

#### 2.2.1. Doctor Agent

In our framework, the decision-making doctor agent’s state space  $s_t \in \mathcal{S}$  encompasses the dialogue history  $H_t$ , providing a comprehensive record of the consultation. The agent’s actions are drawn from the action space  $\mathcal{A} = \{a_{\text{query}}, a_{\text{diagnose}}\}$ , allowing it to generate medical inquiries or execute a diagnostic decision. The environmental dynamics are governed by the patient agent through state transitions, where a new dialogue state  $s_{t+1}$  is determined by the current state  $s_t$  and the action  $a_t$  taken by the doctor agent, as modeled by the transition probability function  $P: s_{t+1} \sim P(s_{t+1}|s_t, a_t)$ . A consultation episode continues until either a predefined turn limit is reached or the doctor agent provides a final diagnosis, at which point a complete dialogue history  $H_T$  forms the trajectory. A reward signal  $R \in \mathbb{R}$  for each trajectory is then provided by an independent consultation evaluator (detailed in Section 2.2.3), considering diagnostic accuracy, information acquisition efficiency, and protocol compliance as key metrics for policy optimization.

To eliminate the requirement for an additional value function approximation, we propose Group Relative Policy Optimization (GRPO) [15] as our policy gradient algorithm. Unlike Proximal Policy Optimization (PPO) [16], GRPO employs the average reward of multiple sampled outputs as a baseline instead of relying on a learned value function. Specifically, for each patient’s consultation  $x$ , GRPO samples a set of trajectories  $y_1, y_2, \dots, y_G$  through the interaction between the doctor agent  $\pi_D$  and the patient agent  $\pi_P$ . The doctor agent, as the policy model, is then optimized by maximizing the following objective function:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}_{\{y_i\}_{i=1}^G \sim \pi_{D, \text{old}}(\cdot|x; \pi_P)} \left[ \frac{1}{G} \sum_{i=1}^G \frac{1}{\sum_{t=1}^{|y_i|} I(y_{i,t})} \sum_{t=1}^{|y_i|} I(y_{i,t}) \right. \\ \left. \min \left( \frac{\pi_D(y_{i,t}|x, y_{i,<t})}{\pi_{D, \text{old}}(y_{i,t}|x, y_{i,<t})} \hat{A}_{i,t}, \text{clip} \left( \frac{\pi_D(y_{i,t}|x, y_{i,<t})}{\pi_{D, \text{old}}(y_{i,t}|x, y_{i,<t})}, 1-\epsilon, 1+\epsilon \right) \hat{A}_{i,t} \right) \right. \\ \left. - \beta \mathbb{D}_{\text{KL}} [\pi_D || \pi_{D, \text{ref}}] \right] \quad (1)$$



**Fig. 2.** The multi-agent collaborative reinforcement learning framework for DoctorAgent-RL. During the rollout stage, multi-turn interactions are conducted between the doctor agent and the patient agent.

where  $\epsilon$  and  $\beta$  are hyperparameters, and  $\hat{A}_{i,t}$  represents the advantage, computed based on the relative rewards of outputs within each group. Here,  $I(y_{i,t}) = 1$  indicates that the token  $y_{i,t}$  is generated by  $\pi_D$ . Since doctors cannot predict patients’ symptoms in advance, during training, the responses of the patient agent are masked.

### 2.2.2. Patient Agent

The patient agent is implemented using Qwen2.5-7B-Instruct [17] and is incorporated into a two-phase dialogue simulation framework via carefully crafted prompt engineering. In the first phase, the system combines patient self-reports and multi-turn dialogue content to create case descriptions. Additionally, it augments potential symptom features using standard diagnostic results, thereby forming a more comprehensive hidden medical profile. This design effectively mitigates the symptom coverage issues stemming from incomplete doctor inquiries in traditional datasets. In the second phase, the patient agent utilizes dynamic symptom release strategies in response to the real-time queries of the doctor agent. It maintains strict pathological consistency while mimicking the natural variability in patients’ symptom description granularity and the order of their complaints. By retaining complete hidden case data, the patient agent ensures that its natural language responses adhere to clinical standards and are generated solely based on the dialogue history.

### 2.2.3. Consultation Evaluator

Our Consultation Evaluator uses a composite reward function designed to train the agent on key clinical skills. This function assesses: 1) Diagnostic Accuracy, measured by the F1 score between the agent’s output and the ground truth diagnosis/recommendation; 2) Information-Gathering Efficiency, which rewards effective questions that elicit information and penalizes repetitive queries; and 3) Protocol Compliance, which penalizes deviations from the required dialogue structure or failing to diagnose within a set number of turns. By integrating these rewards, the agent is optimized not just for correctness but also for conducting a clinically sound and efficient patient interaction. For a detailed implementation, please refer to our code on GitHub.

## 2.3. Training Procedure

As illustrated in Figure 2, our training framework for the doctor agent is built upon Qwen2.5-7B-Instruct, following the DeepSeek-R1 training paradigm [18]. The approach employs a two-stage training pipeline, integrating SFT and RL to cultivate clinical reasoning capabilities.

Specifically, we randomly select 1,000 multi-turn consultation dialogues from the training corpus. Each doctor’s query in the sampled dialogues is augmented with structured thought processes using DeepSeek-V3. These include hypothesis generation, evidence evaluation, and differential diagnosis steps derived from context. The doctor agent is fine-tuned on this enriched dataset to activate core capabilities in reasoning, question-asking, and diagnostic generation.

Following SFT, we refine the agent’s decision-making under interaction constraints using the policy optimization algorithm detailed in the Task Formulation section. To enhance robustness and mimic real-world clinical scenarios, we introduce a **Dynamic Turn Budget Training Strategy**. Each training episode is assigned a random dialogue turn budget (2–10 turns). The model is explicitly reminded of the remaining turns after each interaction step, encouraging efficient information gathering. This two-stage approach ensures the agent first internalizes clinical reasoning patterns via supervised learning, then refines its strategy through interactive reward optimization.

## 3. EXPERIMENT

### 3.1. Comparative Performance Evaluation of Multi-Model Approaches on MTMedDialog

Evaluation metrics include diagnostic accuracy, assessed by an LLM judge on a 6-level scale (0-5) against the gold standard and converted to a 100-point scale (score  $\times$  20), and the average number of dialogue turns. Experimental results in Table 1 show DoctorAgent-RL achieving a comprehensive average score of 53.9%, demonstrating significant improvements over frontier models, open-source base models, and domain-specific models. The system particularly maintains stable advantages in disease types requiring in-depth consultation, proving its superior ability to simulate doctors’ clinical consultation processes.

Analysis of model dialogue processes reveals key limitations in existing methods’ interaction quality. While all models receive explicit instructions to “ask only one question at a time,” frontier models show strict compliance but lack professional medical consultation knowledge, often missing critical symptom information in their questions. Some open-source models (e.g., GLM-4, Mistral) achieve decent comprehensive average scores but improperly combine multiple questions, reflecting inadequate instruction-following capability that affects complex condition diagnosis. The domain-specific model BioMistral exhibits the highest interaction fre-

| Model                          | DSD         | RSD         | ID          | GSD         | ND          | CSD         | ED          | SD          | Avg. Score  | Avg. Turns |
|--------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------|
| <i>Frontier Models</i>         |             |             |             |             |             |             |             |             |             |            |
| GPT-4o                         | <u>49.5</u> | <u>50.7</u> | <u>48.6</u> | <u>46.8</u> | <u>47.9</u> | <u>52.5</u> | <u>46.3</u> | 41.5        | <u>49.4</u> | 3.8        |
| DeepSeek-V3                    | 44.8        | 46.6        | 44.2        | 45.0        | 45.1        | 46.3        | 43.7        | 43.0        | 45.2        | 3.3        |
| LLaMA-3.1-70B                  | 45.4        | 44.5        | 45.6        | 46.1        | 42.1        | 44.9        | 39.5        | 39.0        | 45.0        | 6.0        |
| <i>Open-Source Base Models</i> |             |             |             |             |             |             |             |             |             |            |
| GLM-4-9B                       | 37.5        | 38.3        | 39.7        | 35.2        | 39.1        | 38.3        | 40.0        | 30.5        | 37.5        | 2.1        |
| LLaMA-3.1-8B                   | 38.8        | 38.7        | 39.3        | 37.6        | 39.0        | 39.6        | 36.3        | 41.5        | 38.8        | 4.8        |
| Mistral-7B-Instruct            | 40.3        | 41.5        | 39.8        | 41.9        | 44.2        | 39.8        | 36.3        | 35.5        | 40.6        | 3.4        |
| Qwen2.5-7B-Instruct            | 43.2        | 42.2        | 40.8        | 43.6        | 39.5        | 45.5        | 35.0        | 40.5        | 42.6        | 6.3        |
| <i>Domain-Specific Models</i>  |             |             |             |             |             |             |             |             |             |            |
| BioMistral                     | 28.8        | 28.9        | 29.2        | 23.9        | 29.5        | 25.6        | 37.0        | 25.5        | 28.6        | <b>9.1</b> |
| UltraMedical-8B                | 42.8        | 43.8        | 39.9        | 41.7        | 44.7        | 45.4        | 42.2        | 38.5        | 42.9        | 3.5        |
| HuatuoGPT-o1-7B                | 43.4        | 44.5        | 44.5        | 41.8        | 43.3        | 49.4        | 45.6        | 37.0        | 43.7        | 2.0        |
| AI Hospital                    | 46.3        | 47.5        | 44.8        | 46.0        | 46.1        | 44.4        | 45.9        | <u>43.5</u> | 46.3        | 3.9        |
| DoctorAgent-RL (Ours)          | <b>54.5</b> | <b>52.9</b> | <b>55.1</b> | <b>52.4</b> | <b>50.4</b> | <b>57.0</b> | <b>51.4</b> | <b>48.0</b> | <b>53.9</b> | <b>8.6</b> |

**Table 1.** Mean diagnostic and recommendation accuracy by disease category on MTMedDialog. DSD (Digestive), RSD (Respiratory), ID (Infectious), GSD (Genitourinary), ND (Neurological), CSD (Circulatory), ED (Endocrine), SD (Skin). Best performing metrics are highlighted in bold. The second best results are indicated with underlines.

| Method                | DSD         | RSD         | ID          | GSD         | ND          | CSD         | ED          | SD          | Avg. Score  | Avg. Turns |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------|
| DoctorAgent-RL (Ours) | <b>54.5</b> | <b>52.9</b> | <b>55.1</b> | <b>52.4</b> | 50.4        | <b>57.0</b> | <b>51.4</b> | <b>48.0</b> | <b>53.9</b> | 8.6        |
| w/o Dynamic Turn      | 53.5        | 52.4        | 51.4        | 51.4        | <b>50.9</b> | 51.8        | 44.1        | 45.0        | 52.7        | 8.5        |
| w/o SFT               | 48.4        | 48.3        | 48.4        | 48.8        | 49.5        | 50.0        | 48.2        | 43.0        | 48.4        | 5.8        |
| w/o RL                | 48.3        | 46.5        | 43.6        | 44.5        | 49.0        | 50.2        | 48.5        | 36.0        | 47.4        | <b>9.0</b> |
| Qwen2.5-7B-Instruct   | 43.2        | 42.2        | 40.8        | 43.6        | 39.5        | 45.5        | 35.0        | 40.5        | 42.6        | 6.3        |

**Table 2.** Performance comparison of different fine-tuning methods for Qwen2.5-7B-Instruct on MTMedDialog. The best results are highlighted in bold.

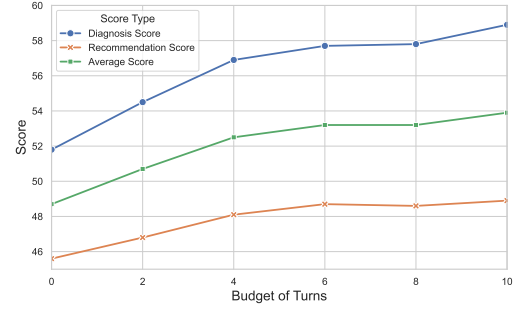
quency but frequently repeats similar questions due to ineffective planning, resulting in the lowest diagnostic accuracy. This highlights that the effectiveness of a diagnosis relies on the quality of questions, not the quantity. DoctorAgent-RL achieves this by dynamically adjusting its inquiries based on patient responses while adhering to consultation protocols, leading to a significantly improved overall score.

### 3.2. Adaptive Fine-Tuning Strategies for Task-Specific Optimization

Experimental results demonstrate the significant advantages of our proposed two-stage optimization framework (SFT + RL) for medical dialogue tasks. As shown in Table 2, DoctorAgent-RL achieves a superior average score of 53.9, outperforming all baseline models. Ablation studies validate the necessity of each component: *w/o Dynamic Turn* causes strategic rigidity during inference with only 1.2% performance drop in matched scenarios; *w/o SFT* leads to 5.5% performance degradation with insufficient questioning initiative; *w/o RL* results in 6.5% lower score with mechanical questioning behavior. The comprehensive performance confirms SFT establishes behavioral baselines, RL enables dynamic decision-making, and adaptive turn mechanisms ensure flexibility—together forming a reproducible paradigm for medical dialogue optimization.

### 3.3. Impact Analysis of Budget of Turns on Diagnosis and Recommendation Performance

The experimental results shown in Figure 3 demonstrate that as the budget of turns increases, the average performance of



**Fig. 3.** Comparative analysis of DoctorAgent-RL performance across budget of turns.

diagnosis and recommendation exhibits a distinct two-phase characteristic: in the initial phase (low-turn range), performance rises rapidly with additional turns, primarily due to the multi-turn dialogue mechanism enabling the LLM to progressively collect and refine patient information through iterative questioning; in the intermediate phase (medium-to-high turn range), the performance curve’s slope noticeably flattens as the valuable information patients can provide gradually becomes saturated, making it difficult for the LLM to extract more meaningful information through additional questioning. Nevertheless, on the whole, a larger budget of turns still leads to better performance, as additional interaction opportunities can capture potential subtle information differences. It is worth noting that diagnostic performance consistently outperforms recommendation performance, as diagnosis tasks can progressively confirm symptom characteristics through multi-turn interactions, while recommendation tasks rely more on established medical knowledge bases, leaving relatively limited room for performance improvement.

## 4. CONCLUSION

In conclusion, we propose DoctorAgent-RL, a multi-agent reinforcement learning framework that addresses the unrealistic requirement of single-turn symptom description and the lack of proactive questioning in static models by modeling clinical consultations as dynamic decision-making under uncertainty. Furthermore, we introduce the MTMedDialog dataset, which features hidden patient profiles enabling progressive symptom revelation, providing a realistic environment for training and evaluating interactive diagnostic agents. Experimental results demonstrate that our approach outperforms existing baselines in both multi-turn reasoning and diagnostic accuracy. This work not only represents a paradigm shift from static Q&A to dynamic reasoning in clinical AI, but also offers practical value by reducing misdiagnosis risks in time-sensitive scenarios, freeing clinicians to handle complex cases, and providing a viable technical pathway for optimizing medical resource allocation.

## 5. ACKNOWLEDGMENTS

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